



Research paper

Fostering early career teachers' preparedness, self-efficacy and professional commitment: The role of teacher education

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ABSTRACT

Retaining high-quality early career teachers (ECTs) has become a considerable challenge in recent years. Few studies have concentrated on the role played by teacher education. Building on Shulman's conceptual framework of the "knowledge-base for teaching," the study investigates its contribution to ECTs' sense of preparedness, self-efficacy, and professional commitment. The results demonstrate a direct positive correlation between the components of the teacher education knowledge base and ECTs' sense of preparedness and self-efficacy. Moreover, the knowledge acquired during teacher preparation indirectly contributes to teachers' professional commitment, mediated by a sense of preparedness and self-efficacy. Implications for theory and practice are discussed.

1. Introduction

Quality education is a crucial factor for the development of individuals, societies, and of economic progress (Akiba & LeTendre, 2018; European Commission, 2018; Hanushek, 2011). As a result, the quality of teachers, those responsible for education delivery, has become a national concern (Goodwin & Low, 2021). There is a broad consensus that teachers are important for students' cognitive, social, and emotional development as well as for their achievement and performance (Carlsson et al., 2019; Goldhaber & Brewer, 2000). It is also a given that supporting student learning and growth requires teachers to be professionally qualified (Carlsson et al., 2019; Donitsa-Schmich & Zuzovsky, 2014), a task in the hands of the teacher education, which is responsible for the recruitment and training of teachers. Although various factors play a role in contributing to teacher quality, teacher education is an important aspect and presents an opportunity to make a difference. Therefore, teacher education merits further attention.

Questions about the effectiveness of teacher education systems have provoked public debate over the years, worldwide (Carmi & Tamir, 2020; Darling-Hammond, 2020; Von Hippel & Bellows, 2018). These questions have led to long-standing demand for empirical research to assess the value of its programs and methods. A large body of knowledge focuses on students' achievements as a key measure of the effectiveness of a training institution (Boyd et al., 2009; Gansle et al., 2012; Lavery et al., 2019). Other factors mentioned in the professional literature are

related to teacher recruitment (DeLuca, 2012; Mihelic et al., 2018) and retention rates (Guarino et al., 2006; Zavelevsky & Shapira Lishchinsky, 2020). Some studies investigate the effectiveness of teacher education through the lens of the academic curriculum, with respect to its content, pedagogy, practical models, evaluation standards, etc. (Cochran-Smith et al., 2015; Darling-Hammond & Bransford, 2005; Orland-Barak & Wang, 2020).

Additional components of teacher education effectiveness are associated with teachers' sense of preparedness, self-efficacy, and professional commitment. These components are subjective measures which represent how well-prepared teachers feel in order to meet the challenges and tasks of the teaching profession (Stites et al., 2018). These factors are consequential since retaining high-quality early career teachers (ECTs) has become a considerable challenge (Ingersoll, 2005; Zavelevsky et al., 2022). There is little empirical evidence that points to the relationship between teachers' sense of preparedness and their actual preparedness to cope with teaching duties and responsibilities; nevertheless, studies indicate that a sense of preparedness is positively associated with fostering positive feelings towards teaching, higher levels of self-efficacy and resilience (Brown, 2021; Giallo & Little, 2003). Professional commitment was also found to promote teachers' development and increase teachers' intentions to stay in the profession (Klassen & Chiu, 2011). While it is evident that a sense of preparedness, self-efficacy, and professional commitment are important motivational constructs with high potential to shape and improve teacher quality and

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teacher retention, there is still much to be learned about how they are developed during the training period.

Building on Shulman’s conceptual framework of a knowledge base for teaching (Shulman, 1986, 1987), the current study analyzes the knowledge domains considered to be main components of teacher education programs and seeks to investigate their contribution to ECTs’ sense of preparedness, self-efficacy, and professional commitment. The study employs a retrospective approach wherein teachers look back and relate their perceptions of the professional preparation they received in their teacher education programs. This approach reflects a reconstructed appraisal of the quality of earlier events (Brewin et al., 1993; Liddicoat & Krasny, 2013). It has been criticized for being based on selective memory and representing a biased evaluation (Gonzales-Morganti et al., 2014). Still, its supporters claim that retrospective evaluation provides an opportunity to generate a deeper understanding of professional knowledge and circumstances from a more mature perspective (Cherubini et al., 2011; Conway, 2013). According to Nasser Abu-Alhija and Fresko (2016), the retrospective process has the potential to identify development paths along the professional career route that may influence teachers’ attitudes toward the teaching profession.

The study focuses on teachers in the first two years of their teaching career and aims to retrospectively explore their perceptions of teacher preparation and the extent to which these perceptions are associated with their evaluations of their sense of preparedness, self-efficacy, and professional commitment. These retrospective perceptions are reflected through teachers’ initial experiences in the education system and may contribute to the study by highlighting the strengths and weaknesses of the interface between teacher education and the field of education and suggest paths for improving teacher preparation and tailoring it to beginning teachers’ needs.

2. Literature review

2.1. Dimensions of knowledge in teacher education

Shulman’s well-documented papers (1986, 1987) describe a conceptual framework of a knowledge base for teaching, while identifying seven distinct domains that represent the main knowledge components required to perform professionally in the classroom and succeed in improving student learning. These dimensions combine content, pedagogy, curriculum, and context, and include the knowledge needed to teach, knowledge needed for teaching, and knowledge of schooling processes. The domains: content knowledge, general pedagogical knowledge, pedagogical content knowledge, curriculum knowledge, knowledge of learners and their characteristics, knowledge of educational context, knowledge of educational ends, purposes, and values are presented and identified in Table 1. According to Herold (2019), these dimensions provide a useful tool to facilitate discussion and analysis of teacher knowledge in different educational contexts, although their interconnections should be further examined.

Shulman’s dimensions of knowledge were widely examined in research on teacher education. The first three dimensions, content knowledge, general pedagogical knowledge, and pedagogical content knowledge have received extensive attention in the professional literature, with the aim of highlighting in depth the challenges of transforming subject matter into content accessible to students (Aydin et al., 2015; Depaepe et al., 2013; Geddis, 1993; Gess-Newsome et al., 2017; Kind & Kennedy, 2019; Krauss et al., 2008; Kunter et al., 2013; Neumann et al., 2019; Sorge et al., 2017). Fewer studies have offered broader exploration of Shulman’s knowledge base conceptualization. Herold’s (2019) study among pre-service teachers in physical education training in England suggested that the balance between Shulman’s dimensions changes in line with the purposes and needs of the different stages of training.

The current study takes Shulman’s work a step further while addressing its relevance during the induction period when teachers are

Table 1
The knowledge base of teaching categories by Shulman, 1987.

Domains of knowledge	
Content knowledge	The content and structure of specific subject matter.
General pedagogical knowledge	The principles, strategies, and methods of instruction and learning, classroom management and organization which transcend specific subject matter.
Pedagogical content knowledge	The integration of content and pedagogy which facilitates organization of subject matter or general topics for instruction in a way that makes it comprehensible to others.
Curriculum knowledge	Teachers’ understanding of the programs, topics, and instructional materials included in and designed for teaching a particular subject.
Knowledge of learners and their characteristics	Understanding the cognitive, physical, psychological, social, personal, and cultural aspects which influence students’ needs and interests.
Knowledge of educational context	Becoming familiar with the social and cultural dynamics of school and community.
Knowledge of educational ends, purposes, and values	The philosophical ideas, historical grounds, broader goals and values of education and schools.

expected to convey the knowledge they acquired throughout their training to the students and at the same time, cope with the challenges inherent in the process of assimilating into the education system. Therefore, we addressed domains of knowledge not considered in Shulman’s original framework, but which are fundamental contextual factors in the stage of entrance into the education system (Courtney et al., 2023): knowledge of classroom management, school organization, parental involvement, and knowledge about professionalism in teaching. In Shulman’s original conceptualization, knowledge of classroom management was initially embedded within the general pedagogical domain. In the current study, we suggest it be recognized separately, due to the significant concerns it presents to ECTs in the first years of teaching (Dicke et al., 2015; Voss et al., 2017). In this manner, general pedagogical knowledge refers to instructional methods, while classroom management describes practices that support dealing with student heterogeneity, individual needs, and discipline (Doyle, 2013). The knowledge of school organization refers to the structure and work processes in the educational institutions. Knowledge of parental involvement refers to practices which support coping and communicating with parents (Addi-Raccah and Arviv-Elyashiv, 2008). The final domain, knowledge of teaching professionalism, refers to the components upon which the foundations of the profession are built (such as: values, responsibilities, and ethics) and which foster professional identity.

The study examines how ECTs in the induction period, the first and second years of their career, perceive the relationship between the development of knowledge for teaching and their sense of preparedness, their self-efficacy to perform as teachers, and their professional commitment. The following sections will elaborate on these issues.

2.2. Sense of preparedness and self-efficacy

The concept of preparedness is connected to the theoretical framework of social-cognitive theory, and more specifically, to Bandura’s work on self-efficacy. Self-efficacy is defined as an individual’s belief in his or her capabilities to reach certain goals and gain given attainments (Bandura, 1977, 1997). Teachers’ self-efficacy demonstrates the teachers’ sense of ability to efficiently accomplish their responsibilities and positively contribute to student learning (Tschannen-Moran & Woolfolk-Hoy, 2007). Previous studies showed that efficacious teachers have extensive content knowledge, feel they possess pedagogical expertise (Klassen & Chiu, 2010), and are confident they can address educational objectives, meet the challenges of the teaching profession,

and contribute to student performance (Katz & Stupel, 2016; Tsouloupas et al., 2014).

Teachers' self-efficacy is related to another conceptual cognitive aspect, known in the educational research literature as teachers' sense of preparedness (Giallo & Little, 2003; Stites et al., 2018). This notion builds on the concept of readiness, and it does not refer to individuals' sense of ability, as does self-efficacy. Sense of preparedness represents the confidence of being well-prepared to execute the duties and responsibilities of teaching (Housego, 1990). Wright's (2017) examination, for example, noted that first-year teachers ($n = 144$) and school principals ($n = 167$) reported that the former were well prepared in the areas of content knowledge, instruction, curriculum, assessment, and professionalism. However, they were less prepared to deal with classroom management. Kee's (2012) study among 1690 first-year teachers indicated that sense of preparedness was related to the development of pedagogical skills via didactic courses and clinical experience attained during teacher preparation. Likewise, Brown et al. (2015) found that practical teaching experience and observation of experienced teachers teaching classes contributed highly to the establishment of a sense of preparedness among pre-service teachers. Darling-Hammond's comparison of teacher education practices in five countries (Darling-Hammond, 2017) showed similar results.

The relationship between teacher preparation and ECTs' sense of preparedness and self-efficacy is backed by some evidence in the professional literature (Brown et al., 2015, 2021; Klassen & Chiu, 2011; Tschannen-Moran & Woolfolk-Hoy, 2007). Yet, these examinations were indirect and focused on contextual factors. This study contributes to the professional literature by addressing the various components of knowledge acquired in teacher education, and by directly examining their relationship to ECTs' sense of preparedness and self-efficacy.

Considering the above literature review, the study will examine the following hypotheses.

H1. The knowledge base components acquired in teacher education are positively correlated with ECTs' sense of preparedness.

H2. The knowledge base components acquired in teacher education are positively correlated with ECTs' self-efficacy.

2.3. Professional commitment

Professional commitment is a psychological state that reflects attitudes towards a chosen career. It demonstrates a bond in which the beliefs and goals of the individual become increasingly integrated with the values and norms of the profession, involvement with its development and progress, and loyalty to the profession's membership (Lam, 2011; Morrow & Wirth, 1989). Teachers' professional commitment is reflected in emotional attachment to teaching, encouraging career development and dedication to remaining in the profession for a long-term career (Meyer & Allen, 1991). Studies have shown that teachers who are committed to the teaching profession identify with its values and ideas, feel satisfied and motivated (Arviv Elyashiv & Gal, 2017; Shen et al., 2012) as well as express confidence in their career choice, and the intention to remain in their teaching position (Kelchtermans, 2017).

Professional commitment has received increasing, but still limited, attention in the teaching arena. For instance, Yang et al. (2019) tested a three-level model to examine the effects of teacher characteristics, school environment, and district level human resource management (HRM) practices on teacher commitment. The results showed that multiple sources of interpersonal support and collegial relationships were the most consistent and strongest variables in predicting teacher commitment. The role of teacher education in determining teachers' commitment and intentions to stay in the teaching profession should be further explored. Therefore, another goal of this study was to investigate the relationship between teacher preparation and ECTs' professional commitment. It is probable that this relationship is indirect and affected

by the variety of experiences during the initial stage of the teaching career. Self-efficacy and a sense of preparedness play major roles in predicting teachers' attitudes toward teaching (Brown et al., 2015; Reeves et al., 2022) and thus, we assume they may serve as mediators in this context. Accordingly, the hypotheses below test these relationships.

H3. The knowledge base for teaching (H3.1), sense of preparedness (H3.2) and self-efficacy (H3.3) are positively correlated with teachers' professional commitment.

H4. Teachers' teaching preparedness and self-efficacy are mediator variables between the knowledge base for teaching and teachers' professional commitment.

2.4. Context of the study

The study investigated these issues in the Israeli education context. ECTs worldwide report feeling unprepared to deal with the professional demands of their job (Reeves et al., 2022); many of them (30%–50%) decide to leave the profession (Papay et al., 2017; Ryan et al., 2017). Likewise, ECTs in Israel consistently feel uncertain about their ability to succeed in teaching effectively (Carmel & Badash, 2018). Moreover, within the first five years of teaching about one-third of ECTs leave the teaching profession, most of them during the first year (13%–16%) (ICBS, 2018). A lower proportion of attrition, but still rather high, occurs during the second year of teaching (5%–7%) (Arviv Elyashiv & Navon, 2021). Similarly, it is interesting to investigate whether there are parallel differences between first- and second-year teachers during the induction period, with respect to other dimensions: sense of preparedness, self-efficacy, and professional commitment. We assume that more teaching experience in the second year is correlated with higher levels of preparedness, self-efficacy, and commitment. The current study examines this hypothesis.

H5. Second-year teachers will show higher levels of preparedness, self-efficacy, and professional commitment than first-year teachers.

2.5. Study aim

The purpose of this study is to explore ECTs' perceptions of teacher preparation and the extent to which these perceptions are associated with their sense of preparedness, self-efficacy, and professional commitment, while examining the differences between first- and second-year teachers. Fig. 1 summarizes the study model.

3. Methods

3.1. Participants

Data was collected at the largest teachers' education college in Israel, located in the country's central region, during the course of induction workshops for ECTs. The college trains teachers in a variety of tracks, such as: sciences, the humanities, special education, the arts, and

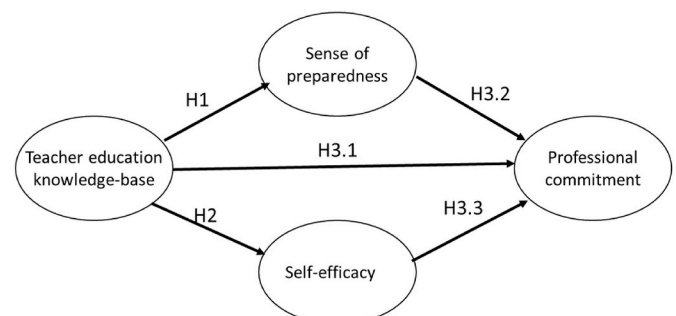


Fig. 1. The study model.

physical education. An average of 800 first-year teachers and 400 s-year teachers participate in these workshops. The participants are representative of teachers in the secular Jewish sector in Israel. We sent an invitation to participate in the study to 794 first-year teachers and 412 s-year teachers, who participated in the induction workshops for ECTs in 2018. Of these, 417 (53%) first-year teachers and 167 (41%) second-year teachers accepted our invitation – a total of 584 ECTs (83.4% women, average age of 31.66 years, $SD = 6.57$). Among them, 124 participants (21.2%) taught kindergarten, 261 participants (41.7%) taught in primary schools and 193 participants (33%) taught in secondary schools (six participants did not respond to this question). Moreover, 397 participants (68%) graduated from traditional tracks, while the other 187 participants (32%) studied in fast-track teacher training programs. The average number of working (teaching) hours was 23.06 weekly hours ($SD = 8.6$). The study received ethics approval from the Institute Review Board for the Protection of Human Participants in Research.

3.2. Research instrument

A questionnaire to be answered anonymously was prepared as a part of an extensive study, conducted from 2018 to 2020, which aimed to explore the relationship between teacher education and ECTs' experiences in the education system. It was prepared using the following validated pre-existing questionnaires: teaching retention (Arviv Elyashiv & Zimmerman, 2013), teacher quality (Levi-Feldman, 2010), and teachers' professional identity (Fisherman & Weiss, 2011). The first version of the questionnaire was examined by three professional experts in the field of teacher education. After some improvements, a second version was completed by two groups of ECTs participating in an academic workshop. Their comments and suggestions were documented. The revised questionnaire was further examined by another professional expert.

The final research questionnaire for the current study included the following sections. The first section was used to obtain background information. The second (37 items) referred to the components of the knowledge base acquired in teacher education (Shulman, 1986, 1987). The third section (5 items) examined teachers' sense of preparedness, the fourth (19 items) explored teachers' self-efficacy, and the fifth (4 items) examined teachers' professional commitment. All items were measured on a Likert scale, ranging from 1 ("do not agree") to 5 ("strongly agree").

3.3. Variables

1. Knowledge base for teaching: This concept consists of nine components based on Shulman's model, which were extracted and constructed using the factor method of principal components analysis (Kim & Mueller, 1978).

- Content knowledge (5 items, $\alpha = 0.88$), representing the content and structure of the subject matter.
- Pedagogical knowledge (6 items, $\alpha = 0.90$), representing the variety of instruction methods and strategies.
- Pedagogical content knowledge (6 items, $\alpha = 0.89$), representing the integration of content and pedagogy which facilitates organization of subject matter for instruction.
- Classroom management (6 items, $\alpha = 0.88$), representing practices which encourage dealing with students' heterogeneity, individual needs, and discipline.
- School organization (4 items, $\alpha = 0.90$), representing the structure and work processes at the educational institutions.
- School curriculum (2 items, $\alpha = 0.88$, $r = 0.60$), representing familiarity with the official curriculum.
- Social diversity (3 items, $\alpha = 0.89$), representing familiarity with the social diversity of students and community.
- Parental involvement (1 item)

- Teaching professionalism (4 items, $\alpha = 0.85$), representing the components that form the foundations of the profession and foster professional identity.

2. Sense of preparedness: Five items representing ECTs' sense of being well-prepared to function professionally as teachers ($\alpha = 0.90$).

3. Self-efficacy: Nineteen items representing ECTs' sense of their ability to professionally perform as teachers ($\alpha = 0.96$).

4. Professional commitment: Four items representing ECTs' motivation to develop professionally and remain in the education system ($\alpha = 0.85$).

The list of the items is presented in Appendix A.

3.4. Control variables

- Years of teaching:** 1. first-year; 2. second-year.
- Working hours** (per week)

3.5. Data analysis

Data were analyzed using SPSS version 29, and includes descriptive statistics procedures, independent t-tests, and Pearson correlations. A linear regression was used to estimate the effect of the components of the knowledge base for teaching on teachers' sense of preparedness, self-efficacy, and professional commitment. To assess the mediation relationships between the study variables, AMOS version 26 was used to conduct a path analysis in Structural Equation Modeling (SEM) using a bootstrapping method. The bootstrapping analysis was based on 5000 iterations to assess the estimators, using 95% confidence intervals for direct and indirect effects. Six indices were estimated to evaluate the model. The first measure was chi-squared, which is satisfactory when the value is not significant. Three measures: the goodness of fit index (GFI), the comparative fit index (CFI), and the non-normed fit index (NFI) are considered adequate when values are above 0.90, while excellent fit is above 0.95. The last two measures: the root mean square error of approximation (RMSEA) and the standardized root mean square residual (SRMR), are considered adequate values when they are less than 0.08, while excellent fit is less than 0.06 (Arbuckle, 2013). The level of significance (p-value) was 5%.

4. Results

Descriptive statistics and correlations of the study variables are presented in Table 2. The correlation matrix revealed that the components of the knowledge base for teaching, sense of preparedness, self-efficacy, and professional commitment among ECTs were positively correlated. An examination of the means of the study variables indicated that ECTs feel highly committed to the teaching profession ($M = 3.83$, $SD = 0.87$). However, they revealed a moderate level of preparedness ($M = 3.07$, $SD = 0.92$) and self-efficacy ($M = 3.14$, $SD = 0.83$). Similarly, teachers conceded that the knowledge they acquired during teacher preparation was hardly sufficient.

Fig. 2 highlights the differences between first-year teachers and second-year teachers. As shown, first-year teachers feel more prepared to meet the responsibilities and duties of the teaching profession ($t = 5.53$, $p < 0.001$) and possess a higher level of self-efficacy to actively perform as teachers ($t = 5.72$, $p < 0.001$), in comparison to second-year teachers. Non-significant results were found regarding teachers' professional commitment ($t = 1.67$, $p > 0.05$); both first and second-year teachers express a high level of commitment to the teaching profession.

4.1. Sense of preparedness

A linear regression was performed to examine the relationship between the knowledge base components acquired in teacher education and teachers' sense of preparedness, while controlling for teachers'

Table 2
Descriptive statistics and correlations.

	Mean (S.D.)	<u>2</u>	<u>3</u>	<u>4</u>	<u>5</u>	<u>6</u>	<u>7</u>	<u>8</u>	<u>9</u>	<u>10</u>	<u>11</u>	<u>12</u>
1. Professional commitment	3.83 (0.67)	0.21 ^a	0.20 ^a	0.16 ^a	0.14 ^a	0.15 ^a	0.16 ^a	0.14 ^a	0.14 ^a	0.13 ^a	0.11 ^a	0.15 ^a
2. Self-efficacy	3.14 (0.83)	–	0.70 ^a	0.45 ^a	0.63 ^a	0.66 ^a	0.69 ^a	0.60 ^a	0.53 ^a	0.55 ^a	0.47 ^a	0.65 ^a
3. Sense of preparedness	3.07 (0.92)		–	0.56 ^a	0.62 ^a	0.66 ^a	0.65 ^a	0.61 ^a	0.52 ^a	0.48 ^a	0.44 ^a	0.65 ^a
4. Content knowledge	3.46 (0.84)			–	0.45 ^a	0.58 ^a	0.49 ^a	0.38 ^a	0.40 ^a	0.47 ^a	0.28 ^a	0.41 ^a
5. Pedagogical knowledge	3.44 (0.83)				–	0.73 ^a	0.68 ^a	0.49 ^a	0.49 ^a	0.46 ^a	0.30 ^a	0.51 ^a
6. Pedagogical content knowledge	3.21 (0.87)					–	0.73 ^a	0.51 ^a	0.54 ^a	0.53 ^a	0.37 ^a	0.57 ^a
7. Classroom management knowledge	2.85 (0.95)						–	0.59 ^a	0.52 ^a	0.61 ^a	0.49 ^a	0.63 ^a
8. Knowledge of school organization	2.44 (0.96)							–	0.64 ^a	0.50 ^a	0.56 ^a	0.76 ^a
9. Knowledge of curriculum	2.77 (1.04)								–	0.52 ^a	0.43 ^a	0.58 ^a
10. Social diversity knowledge	2.86 (0.94)									–	0.46 ^a	0.54 ^a
11. Knowledge of parental involvement	2.42 (1.20)										–	0.58 ^a
12. Knowledge of teaching professionalism	2.89 (0.96)											–

* $p < 0.05$.

** $p < 0.01$.

^a $p < .001$.

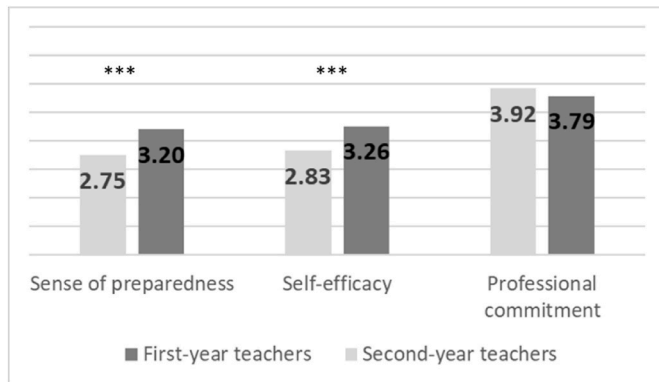


Fig. 2. Differences between first-year and second-year teachers (** $p < 0.001$).

teaching characteristics (years of teaching, weekly hours). The results are presented in Table 3. Six of the knowledge domains (out of nine) were found to be positively associated with ECTs' sense of preparedness. The most dominant component found to contribute to their preparedness was the content knowledge module ($\beta = 0.22$). The second major predictor was knowledge of professionalism in teaching ($\beta = 0.20$). Teachers also perceived pedagogical knowledge ($\beta = 0.16$), pedagogical content knowledge ($\beta = 0.14$), school organization ($\beta = 0.13$), and classroom management knowledge ($\beta = 0.11$) as contributors to a greater sense of preparedness for teaching. Three knowledge base components were not significant in this regard: curriculum, social

Table 3
Linear regression coefficients predicting teachers' preparedness.

	β (S.E)	β
Components of Knowledge		
Content knowledge	0.24(0.04)	0.22 ^c
Pedagogical knowledge	0.17(0.04)	0.16 ^c
Pedagogical content knowledge	0.14(0.05)	0.14 ^b
Classroom management	0.11(0.04)	0.11 ^a
School organization	0.11(0.04)	0.12 ^b
Curriculum	0.02(0.03)	0.03
Social diversity	−0.05(0.03)	−0.05
Parental involvement	0.04(0.03)	0.05
Teaching professionalism	0.19(0.04)	0.20 ^c
Years of teaching (first vs. second year)	0.16(0.06)	0.08 ^b
Weekly working hours	−0.01(0.003)	−0.07 ^c
Intercept	0.09(0.15)	
R ²	0.63	

^a $p < .05$.

^b $p < .01$.

^c $p < .001$.

diversity, and parental involvement.

Another interesting finding is the significant relationship between the “years of teaching” control variable and preparedness. The positive coefficient indicates that first-year teachers report a higher level of preparedness in comparison to second-year teachers. Moreover, among ECTs, a higher number of teaching hours is associated with feelings of lesser preparedness to perform as teachers.

4.2. Self-efficacy

Another linear regression was estimated to examine the relationships between the knowledge base components acquired in teacher education and teachers' self-efficacy. As shown in Table 4, six of the knowledge base components (out of nine) were found to be positively associated with ECTs' self-efficacy. The most dominant component was the knowledge regarding classroom management ($\beta = 0.19$). Acquiring professional knowledge in other areas, such as pedagogical knowledge ($\beta = 0.15$), pedagogical content knowledge ($\beta = 0.16$), knowledge regarding social diversity ($\beta = 0.09$), and teaching professionalism ($\beta = 0.17$) also contributed to the ability of ECTs to perform in the classroom. The years of teaching variable was found to be positively significant, meaning that controlling for the knowledge base acquired in teacher preparation training, first-year teachers feel a higher level of self-efficacy, in comparison to second-year teachers. The coefficient of the other control variable, weekly working hours, had a non-significant effect.

Table 4
Linear regression coefficients predicting teachers' self-efficacy.

	β (S.E)	β
Components of Knowledge		
Content knowledge	0.02(0.03)	0.02
Pedagogical knowledge	0.15(0.04)	0.15 ^c
Pedagogical content knowledge	0.15(0.04)	0.16 ^c
Classroom management	0.16(0.04)	0.19 ^c
School organization	0.07(0.04)	0.08
Curriculum	0.03(0.03)	0.04
Social diversity	0.08(0.03)	0.09 ^a
Parental involvement	0.05(0.02)	0.07
Teaching professionalism	0.15(0.04)	0.17 ^c
Years of teaching (first vs. second year)	0.16(0.05)	0.08 ^b
Weekly working hours	−0.003(0.003)	−0.03
Intercept	0.52(0.13)	
R ²	0.62	

^a $p < .05$.

^b $p < .01$.

^c $p < .001$.

4.3. Professional commitment

Table 5 presents the two-stage linear regression results which examined professional commitment predictors. The first step included only the knowledge base components acquired in teacher education. The coefficients were not significant, meaning that teacher education was barely associated with ECTs' professional commitment. In the second step, sense of preparedness and self-efficacy were included. These variables were shown to increase new teachers' intentions of remaining in the teaching profession. The coefficient of years of teaching was found to be significant. Its negative direction implies that second-year teachers feel a higher level of professional commitment than first-year teachers. It is also apparent that a higher number of working hours is associated with an increase in ECTs' professional commitment.

4.4. Mediating effects

To examine in more depth the relationships between the knowledge base components acquired in teacher education and ECTs' sense of preparedness, self-efficacy, and professional commitment, structural equation modeling (SEM) was performed (Fig. 3). The knowledge base components were used as exogenous variables, ECTs' sense of preparedness and self-efficacy served as mediators, and professional commitment was the outcome tested. In addition, we controlled for years of teaching and weekly working hours. This method enables examination of the direct and indirect effects of the teacher education knowledge base on professional commitment.

In this path model, we decided to use a latent variable, which was constructed by including all of the knowledge base components and provides a general assessment of the teacher education knowledge base ($\alpha = 0.90$) while exploring its effect on the relationship between the mediators and outcome variables. This model provided relatively good goodness of fit indices ($\chi^2(49) = 82.53, p < 0.01$; GFI = 0.98; NFI = 0.98; CFI = 0.99; RMSEA = 0.03; SRMR = 0.03).

The results shown in Fig. 3 demonstrate the direct effects between the study variables. Accordingly, the knowledge base latent variable was positively associated with sense of preparedness ($\beta = 0.82$, S.E. = 0.07, p

< 0.001) and self-efficacy ($\beta = 0.81$, S.E. = 0.06, $p < 0.001$), but had a non-significant effect on ECTs' professional commitment ($\beta = -0.08$, S.E. = 0.15, $p > 0.05$). Still, both sense of preparedness ($\beta = 0.19$, S.E. = 0.07, $p < 0.05$) and self-efficacy ($\beta = 0.19$, S.E. = 0.08, $p < 0.05$) were positively correlated with professional commitment. These results further confirm the linear regression results mentioned above.

Table 6 provides an overview of the direct and indirect effects of the study variables on ECTs' professional commitment. The results indicate that the indirect effect of the knowledge base for teaching on professional commitment was found statistically significant, meaning that the perception of teacher education as contributing highly to acquiring the knowledge needed to professionally perform as teachers leads to stronger feelings of professional commitment through the mediation of sense of preparedness and self-efficacy.

The results shown in Fig. 3 also indicate that years of teaching had a positive direct effect on ECTs' sense of preparedness and self-efficacy, while having a negative direct effect on teachers' professional commitment. Weekly working hours had a positive direct effect on ECTs' professional commitment.

5. Discussion

The purpose of this study was to examine ECTs' perceptions of teacher preparation, and the extent to which these perceptions are associated with their sense of preparedness, self-efficacy, and professional commitment. The importance of this research stems from the concern with establishing quality education, a goal which may be impeded by the high attrition rates among ECTs in many countries (Papay et al., 2017; Ryan et al., 2017), including Israel (Arviv Elyashiv & Navon, 2021). The study employed a retrospective approach wherein teachers looked back at their earlier experiences during the teacher preparation process and reflected on its contribution to their current experiences in the field of education (Brewin et al., 1993; Liddicoat & Krasny, 2013). Following Shulman's original knowledge base for teaching conceptual framework (1986, 1987), the study examined the relationship between teacher education knowledge acquired and ECTs' competence and professionalism during the induction stage.

5.1. Sense of preparedness and self-efficacy

The findings of this study indicate that ECTs in the first two years of their career retrospectively perceive the teacher education body of knowledge as beneficial to their teaching experience in terms of sense of preparedness and self-efficacy. Indeed, the results demonstrate a direct positive correlation between the teaching education knowledge base and ECTs' sense of preparedness and self-efficacy. These findings confirm Hypothesis 1 and Hypothesis 2. In order to achieve a deeper understanding of the contribution of teacher education to ECTs' experience during the first years as teachers, the results concerning some knowledge base components warrant closer attention.

First, both pedagogical components: general pedagogical knowledge and pedagogical-content knowledge positively contributed to ECTs' sense of preparedness and self-efficacy. These findings align with previous studies addressing the contribution of studying classroom practices to fostering preparedness and efficacy during the first steps in teaching (Clark et al., 2015; Kee, 2012; Wright, 2017). This body of knowledge includes didactic training and practicum experience (Darling-Hammond, 2010). It aims to instruct teachers on how to make knowledgeable and consistent professional decisions during the course of educational practice (Barrett & Hordern, 2021) and therefore, as shown in the study results, its value to ECTs was observed.

As for content knowledge, it was found to be the most dominant factor in predicting ECTs' sense of preparedness. Subject matter is often perceived as secondary to pedagogical studies and the practicum experience (Pearson, 2016). Still, it establishes the core of professional expertise (Berliner, 2001) and it is essential for optimal teaching

Table 5
Linear regression coefficients predicting teachers' professional commitment.

	Step 1		Step 2	
	B(S.E)	β	B(S.E)	β
Components of Knowledge				
Content knowledge	0.07(0.05)	0.07	0.03(0.05)	0.03
Pedagogical knowledge	0.03(0.07)	0.03	-0.03(0.07)	-0.03
Pedagogical content knowledge	-0.004(0.07)	-0.004	-0.06(0.07)	-0.06
Classroom management	0.07(0.07)	0.07	0.02(0.07)	0.02
School organization	0.04(0.06)	0.05	0.01(0.06)	0.01
Curriculum	0.04(0.05)	0.04	0.03(0.05)	0.03
Social diversity	-0.02(0.05)	-0.02	-0.02(0.05)	-0.03
Parental involvement	-0.003(0.04)	0.01	-0.02(0.04)	-0.03
Teaching professionalism	0.04(0.06)	0.05	-0.02(0.06)	-0.02
Sense of preparedness	-	-	0.16(0.06)	0.16 ^a
Self-efficacy	-	-	0.19(0.07)	0.18 ^b
Years of teaching (first vs. second year)	-0.09(0.08)	-0.08 ^b	-0.15(0.08)	-0.08 ^b
Weekly working hours	0.02(0.003)	0.19 ^c	0.02(0.004)	0.21 ^c
Intercept	2.63(0.22)		2.63(0.22)	
R ²	0.08		0.11	

^a $p < .05$.

^b $p < .01$.

^c $p < .001$.

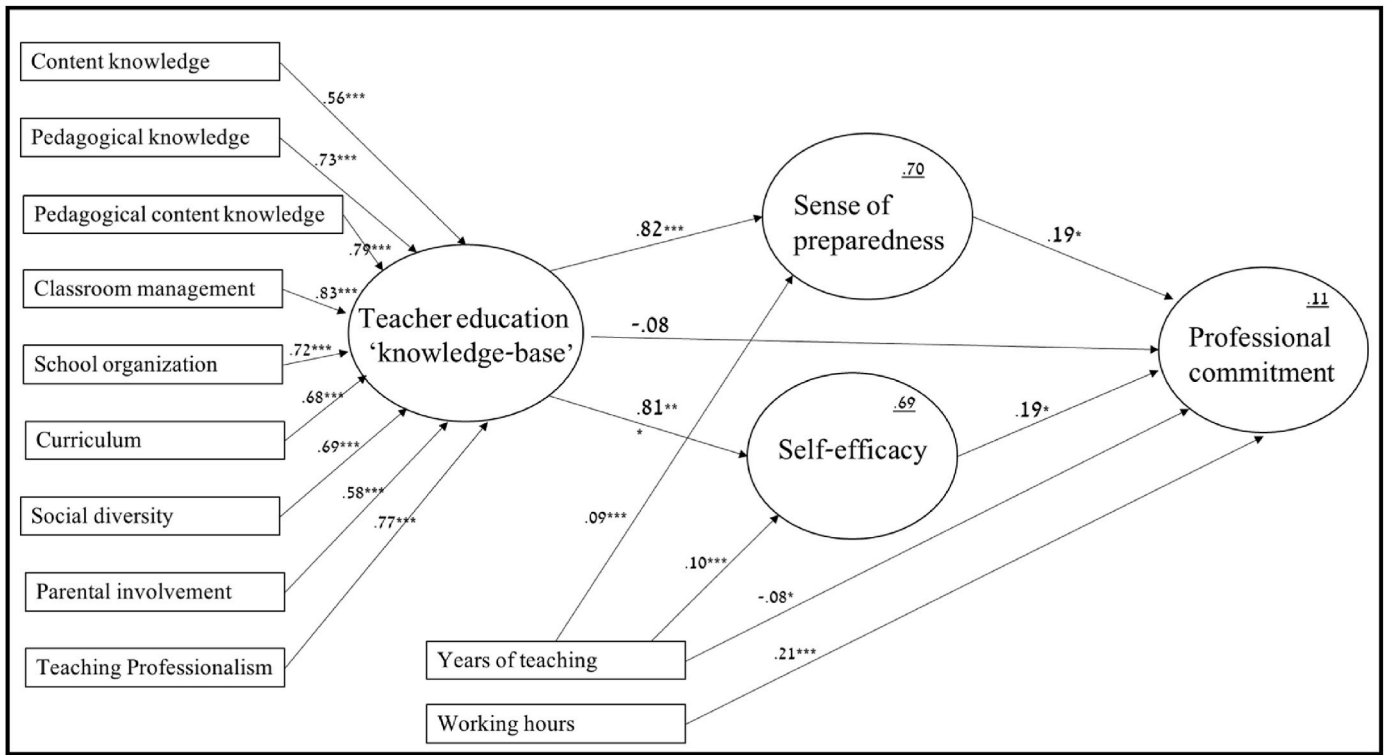


Fig. 3. Structural equation model for the path predicting professional commitment.

Table 6
Direct and indirect effects of study variables on ECTs' professional commitment.

		Direct effect	Indirect effect	Total effect
Teacher education knowledge base	β (S. E.)	-.08 (0.15)	0.31** (0.10)	0.23 ^a (0.05)
Feelings of preparedness	β (S. E.)	0.19* (0.07)	–	0.19* (0.07)
Self-efficacy	β (S. E.)	0.19* (0.08)	–	0.19* (0.08)

^a $p < .05$.

(Shulman, 1987). Our results confirm these findings. Further reinforcement comes from Morris and Hiebert's (2017) examination among math teachers, showing that deeper understanding of mathematical theory was associated with a higher level of preparedness to teach the subject in the classroom.

Another dominant factor that deserves closer attention is knowledge of classroom management. Classroom management is known to be a complex challenge for ECTs, which can lead to emotional exhaustion and burnout (Dicke et al., 2015). Therefore, it was no surprise that the study results indicated that knowledge about how to manage students' heterogeneity, individual needs, and discipline is valuable in increasing ECTs' sense of preparedness and self-efficacy. An additional knowledge base component worth clarifying is professionalism in teaching. By acquiring this knowledge domain, ECTs can develop the understanding of role expectations and engage in reflective and critical observation of their own teaching, embedded within a process that fosters professional identity (Flores, 2020). Although this was not included in Shulman's original conceptualization (1986, 1987), our study found that this was a factor which strengthened ECTs' sense of preparedness and self-efficacy.

5.2. Professional commitment

The results did not yield significant direct correlations between the

teacher education knowledge base and ECTs' professional commitment. Hence, Hypothesis 3.1, namely, the knowledge base for teaching, was not confirmed. Yet, the results showed that the knowledge acquired during teacher preparation indirectly contributed to teachers' sense of professional commitment through the mediation of sense of preparedness and self-efficacy. The latter significantly increased ECTs' professional commitment. These results confirm both Hypothesis 3.2 and Hypothesis 3.3, namely, that a sense of preparedness and self-efficacy, respectively, both positively correlate with teachers' professional commitment. Hypothesis 4 was fully confirmed, showing that teachers' preparedness and self-efficacy are mediator variables between the knowledge base for teaching and teachers' professional commitment.

Teachers' professional commitment reflects a psychological attachment to the teaching career, which has the potential to motivate teachers to remain in the profession (Arviv Elyashiv & Gal, 2017; Kelchtermans, 2017). This is an important issue with respect to ECTs due to the low retention rates among this group in Israel, as in other countries, which leads to instability in schools, loss of human capital (Sutcher et al., 2019), and has negative effects on student learning (Kini & Podolsky, 2016). Previous studies which discussed teachers' commitment and motivations with respect to the profession focused on organizational or ecological factors (Zavelevsky et al., 2022). The current study highlighted the contribution of teacher education. By fostering a sense of preparedness and self-efficacy, teacher education may encourage retention of teachers, mostly during the initial phases of their career. Introducing student teachers to relevant knowledge base components, as was indicated in the study, may indirectly increase their professional commitment and support retention.

5.3. Years of teaching

Hypothesis 5 addressed the differences between first-year and second-year teachers, predicting that second-year teachers will show higher levels of preparedness, self-efficacy, and professional commitment than first-year teachers. Contrary to the hypothesis, the results

indicated that first-year teachers showed a higher level of preparedness and self-efficacy than second-year teachers. In their second year, teachers become more mature and experienced. Still, their lower evaluations of preparedness and self-efficacy are probably linked to the absorption process in the education system. The challenges and dilemmas characteristic of the initial steps in the profession during the first year of teaching are met with a solid support system at school (such as mentoring) and at training institutions (such as induction workshops). These support pillars decrease in the second year and therefore, may bring about feelings of insecurity and burnout (Skaalvik & Skaalvik, 2011) which may have lesser consequences for teachers' sense of preparedness and self-efficacy.

Despite their lower level of preparedness and self-efficacy, teachers' professional commitment is higher in the second year. These results align with Yu and colleagues' (Yu et al., 2021) findings in which ECTs' professional commitment changes across their training steps, implying that despite the difficulties, during the second year, teachers begin to experience more engagement with teaching. In this context, teacher education should direct efforts to support future teachers' confidence and professionalism while establishing support mechanisms as part of the professional development workshops offered throughout the teaching career. This issue should be further examined via qualitative tools.

6. Conclusions

This study deepens the theoretical perspective related to the effectiveness of teacher education while addressing its contribution to ECTs' integration in the education system. Planning and providing quality teacher education programs that offer a broad understanding of five specific knowledge domains: content, pedagogy, content-pedagogy, classroom management, and teaching professionalism, have the potential to positively contribute to ECTs' sense of preparedness and self-efficacy. In the professional literature, these aspects were found to be valuable for improving student learning (Brown et al., 2021; Katz & Stupel, 2016; Tsouloupas et al., 2014). The current study reinforces their direct contribution to ECTs' professional commitment, and their indirect contribution as a mediator of the effect of teacher education. Based on Shulman's "knowledge base" conceptualization, this study contributes to the professional literature by introducing the knowledge domains that are most beneficial to ECTs' experiences in the first two years of their career. These domains construct the foundations of the professional body of knowledge for teaching (Neumann et al., 2019), which enable teachers to generate educational and cultural themes and make them accessible to students (Barrett & Hordern, 2021). Hence, their value during the initial steps in the education system is unquestionable.

Highlighting the way in which ECTs retrospectively perceive the training they underwent may increase the effectiveness of teacher education through a re-evaluation process which would lead to adapting programs according to the unique needs of teachers in the induction period. In this way, teacher education would contribute to the education system, though the contribution would have an indirect impact, i.e., the increase in ECTs' professional commitment. In turn, it may lower intentions to drop out of the profession and increase teacher retention (Arviv Elyashiv & Gal, 2017; Kelchtermans, 2017).

6.1. Limitations

Several limitations of the study that may cause bias warrant attention. First, the study was conducted in one teacher training institution, and although it is the largest institution for teacher training in Israel, it represents the secular sector only. Second, the study results rely on the participants' retrospective self-reports. Such data may be affected by predisposition and selective memory. However, it also provides authentic responses that describe individuals' perceived reality; thus, the results capture the participants' attitudes and beliefs (McCall, 2013) and demonstrate the individual perspective seen from a subsequent,

experienced point of view (Cherubini et al., 2011) and therefore, may be considered reliable. Moreover, procedural remedies were taken to minimize the effect of potential bias (Podsakoff et al., 2003). A future study that would focus on principals' evaluations of ECTs' preparedness, efficacy, and commitment might introduce another perspective of teachers' engagement in school and allow for comparison with the current study. It is also worth investigating the ethnographic development of ECTs' professional commitment and its implications for teacher education. Additionally, in the initial analyses, we included the student age groups variable (kindergartens, primary schools, and high schools) as a control. The results showed non-significant coefficients. Thus, we decided not to include this variable in the final analysis. Nevertheless, due to its potential as a source of variance, further study should qualitatively explore the similarities and differences between the various age groups which teachers teach.

It is important to note that the comparison between first-year and second-year ECTs in the current study included two different cohorts. Thus, in order to fully understand the difference between one and two years of experience retrospectively, it is worthwhile to establish a longitudinal study that will follow student teachers during the preparation stage to the first and second years of teaching and examine the contribution of teacher education to their sense of preparedness, self-efficacy, and professional commitment.

Data availability

The authors are unable or have chosen not to specify which data has been used.

CRediT authorship contribution statement

Rinat Arviv Elyashiv: Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Katya Rozenberg:** Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

We have no relevant interest(s) to disclose.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.tate.2024.104691>.

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